

A $13/4$ lower coefficient for partite intersecting cover numbers

Research note, 7 October 2026. This note gives complete written proofs. The accompanying Lean file checks selected scalar steps, not the entire hypergraph argument. No mathematical priority claim is made.

1. Definitions and results

A finite hypergraph is **simple** if its edges are distinct nonempty sets. It is **intersecting** if every two distinct edges meet. An **r -partite r -uniform** hypergraph has a specified partition of its vertex set into r parts and each edge contains exactly one vertex in each part. Isolated vertices can be discarded. Write $m = |E(H)|$ and let $\tau(H)$ be the minimum size of a vertex set meeting every edge; the empty hypergraph has cover number zero. Degrees always refer to the currently specified hypergraph. All results below concern finite hypergraphs and integer $r \geq 2$.

Theorem 1 (partite cover inequality). Every simple intersecting r -partite r -uniform hypergraph satisfies

$$m \geq 5\tau(H) - \frac{7}{4}r - 5. \quad (1)$$

Equivalently, $\tau(H) \leq m/5 + 7r/20 + 1$.

Define, whenever the class is nonempty,

$$f(r) = \min\{|E(H)| : H \text{ is simple, intersecting, } r\text{-partite } r\text{-uniform, } \tau(H) \geq r - 1\}.$$

This definition does not assume the intersecting case of Ryser's conjecture, $\tau(H) \leq r - 1$. A counterexample with $\tau(H) = r$ also belongs to the class. Theorem 1 gives

$$f(r) \geq \left\lceil \frac{13}{4}r - 10 \right\rceil. \quad (2)$$

For the hypothetical subclass with $\tau(H) = r$, it gives $m \geq \lceil 13r/4 - 5 \rceil$. These inequalities do not settle Ryser's conjecture or determine $f(r)$.

The proof of Theorem 1 uses the maximum-degree-three cover lemma of Sivashankar [S, Lemma 2]. For completeness its full proof is restated, with attribution, in the appendix. We also prove a bound requiring none of that lemma:

Theorem 2 (independent elementary branch). Under the same hypotheses,

$$m \geq 5\tau(H) - \frac{289}{160}r - \frac{57}{16} - \frac{5}{32r}. \quad (3)$$

Consequently

$$f(r) \geq \left\lceil \frac{511}{160}r - \frac{137}{16} - \frac{5}{32r} \right\rceil. \quad (4)$$

Its leading coefficient is $511/160 = 3.19375$.

The contribution of this note is the partite residual estimate (6) and its combination with the inherited residual estimate (12), after degree-five peeling. The estimate (12) and this peeling framework also appear in [S, Section 4]; they are not claimed as new here. In the sources examined, Aharoni–Barát–Wanless [ABW, Corollary 2.6] prove $f(r) \geq 293r/96 + O(1)$, with a related degree-counting and greedy-cover method. Our coefficient $13/4$ is greater than $293/96$. A finite literature check is not a determination of priority.

2. Degree-five peeling

Put $t = \tau(H)$. Repeatedly select a vertex of current degree at least five, record it, and delete all current edges containing it. Stop when the residual hypergraph J has maximum degree at most four. Let k be the number of recorded vertices and $q = |E(J)|$. Each deletion removes at least five previously undeleted edges. The recorded vertices together with any cover of J cover H . Thus

$$m \geq q + 5k, \quad t \leq k + \tau(J), \quad m \geq q + 5(t - \tau(J)). \quad (5)$$

The last inequality remains valid when $t - \tau(J) < 0$. The residual retains the original parts and uniformity and is intersecting unless empty. For $J = \emptyset$, set $\tau(J) = 0$; all residual bounds below still hold.

3. A partite residual estimate

Lemma 3. If J is intersecting, r -partite, r -uniform and has q edges and maximum degree at most four, then

$$\tau(J) \leq \frac{q}{2} + \frac{1}{2} - \frac{q(q-r-1)}{8r}. \quad (6)$$

Proof. Write x_i for the number of vertices of degree i , $1 \leq i \leq 4$, and put $W = x_3/2 + x_4$. Incidence counting gives

$$x_1 + 2x_2 + 3x_3 + 4x_4 = qr. \quad (7)$$

Every unordered edge pair has at least one common vertex, so

$$\binom{q}{2} \leq x_2 + 3x_3 + 6x_4 = \frac{qr}{2} + 4W - \frac{x_1 + x_3}{2} \leq \frac{qr}{2} + 4W.$$

In particular

$$W \geq \frac{q(q-r-1)}{8}. \quad (8)$$

We do not presume that this right-hand side is positive.

In part c , let a_c, b_c count vertices of degrees three and four. The incident edge sets of distinct vertices of that part are disjoint, since an edge has exactly one vertex in the part. Selecting all these vertices therefore covers exactly $3a_c + 4b_c$ edges. Pair the remaining edges arbitrarily and choose one common vertex per pair; for a singleton, choose any of its vertices. These are vertices of the original J , regardless of its degrees after removing covered edges. This proves

$$\tau(J) \leq a_c + b_c + \left\lceil \frac{q - 3a_c - 4b_c}{2} \right\rceil \leq \frac{q}{2} + \frac{1}{2} - \frac{a_c}{2} - b_c. \quad (9)$$

The saving $a_c/2 + b_c$ has sum W across the r parts, so some part has saving at least W/r . Equations (8)–(9) give (6). For $q = 0$, (6) is simply $0 \leq 1/2$. $\square$

Substituting (6) in (5) yields

$$m \geq 5t - \frac{17q}{8} + \frac{5q(q-1)}{8r} - \frac{5}{2}. \quad (10)$$

This bound uses only the partite structure and elementary counting. The identity

$$\begin{aligned} 5t - \frac{17q}{8} + \frac{5q(q-1)}{8r} - \frac{5}{2} &= 5t - \frac{289r}{160} - \frac{57}{16} - \frac{5}{32r} \\ &\quad + \frac{5}{8r} \left(q - \frac{17r+5}{10} \right)^2 \end{aligned} \quad (11)$$

proves Theorem 2, since $r > 0$. The appendix is not used in this proof.

4. A second residual estimate and the 13/4 coefficient

Lemma 4 (Sivashankar [S, Section 4]). If J is intersecting and r -uniform, with q edges and maximum degree at most four, then

$$4\tau(J) \leq q + r + 4. \quad (12)$$

No partite hypothesis is required for this lemma.

Proof. View edges of J as points and, for every degree-four vertex, form its four-point incidence block. Choose a maximal matching of these blocks, consisting of s disjoint blocks. Their defining vertices cover exactly $4s$ edges. On deleting those edges, no degree-four vertex remains in the residual L : such a vertex would have a block disjoint from the matching and would augment it. Thus $\Delta(L) \leq 3$ and L has $q - 4s$ edges. Appendix Lemma A gives

$$\tau(J) \leq s + \tau(L) \leq s + \frac{q - 4s + r + 4}{4} = \frac{q + r + 4}{4}.$$

This also covers an empty L , with its cover number zero. $\square$

Combining Lemma 4 and (5) gives

$$m \geq 5t - \frac{5}{4}r - \frac{1}{4}q - 5. \quad (13)$$

If $q \leq 2r$, (13) immediately gives (1). If $q \geq 2r$, use (10). Subtracting its value at $q = 2r$ gives the exact factorization

$$\begin{aligned} &\left(5t - \frac{17q}{8} + \frac{5q(q-1)}{8r} - \frac{5}{2} \right) - \left(5t - \frac{7}{4}r - \frac{15}{4} \right) \\ &= \frac{(q-2r)(5q-7r-5)}{8r} \geq 0. \end{aligned} \quad (14)$$

Indeed $5q - 7r - 5 \geq 3r - 5 \geq 1$, since $r \geq 2$. Therefore in this case $m \geq 5t - 7r/4 - 15/4$, which is stronger than (1). This proves Theorem 1 and its consequences. It uses no asymptotic threshold and no edge-colouring theorem. $\square$

5. Boundaries, scope and further questions

For an empty hypergraph both theorems are immediate. Repeated edges may be deleted without changing the cover number, intersection or partite properties. Applying the simple result to the distinct edges gives the same lower bounds for the number of distinct edges, and therefore also for a larger count that includes multiplicities. No bound on the size of the parts is assumed. The proof does not assume linearity (at most one common vertex per edge pair); repeated intersection is permitted.

For small r , the additive constants make (2) weak. In particular this does not replace the repository's separate nineteen-edge necessary bound for a rank-six Ryser counterexample. The new advance here is a uniform leading coefficient for the nonempty extremal class $\tau \geq r - 1$, not an existence result for $\tau = r$.

The possible improvements are concrete. At $q \approx 2r$, the two residual estimates meet at leading order. Raising the coefficient above $13/4$ by this route requires a stronger cover estimate there, or a structural restriction on residuals surviving the peeling process. The slack $(x_1 + x_3)/2$ in the pair count is available when many vertices have those degrees. Nearly sharp configurations would instead concentrate incidence in degrees two and four and distribute their four-blocks across the parts. Determining whether these configurations can coexist with a large cover number is a substantive next problem.

Appendix. The attributed degree-three lemma

Lemma A (Sivashankar [S, Lemma 2]). For an intersecting r -uniform hypergraph L with q edges and maximum degree at most three,

$$4\tau(L) \leq q + r + 4. \quad (15)$$

The proof below restates its maximum-matching argument in our notation; this lemma and its local coloured-graph claim are not claimed as our contribution.

Proof. The empty case is immediate. Fixing one edge and counting its intersections with all other edges gives

$$q - 1 \leq \sum_{v \in e} (d(v) - 1) \leq 2r, \quad q \leq 2r + 1. \quad (16)$$

Treat the q edges as points. Each original vertex defines the block of points containing it, of size at most three. Every point pair is contained in a block. Take a maximum matching of the distinct three-point blocks; let its size be s , its covered points C , and its unmatched points U , with $|U| = u$. Thus $q = 3s + u$. There is no three-block inside U . The defining vertices of the matching, followed by a pairing of U , give a cover of size at most $s + \lceil u/2 \rceil$. It suffices to show

$$r \geq u + s - 2. \quad (17)$$

Put $\alpha = r - u - s + 2$, an integer. If $u = 0$, (16) gives $\alpha \geq (r + 5)/3 > 0$. Assume $u > 0$.

Select one witness vertex for each pair in U . These witnesses are distinct: witnesses for disjoint pairs cannot coincide because the maximum degree is three; witnesses for overlapping pairs cannot coincide because that would create a three-block inside U . Each selected witness contributes exactly two incidences with U ; its block is the selected pair, possibly together with one point of C . The witnesses contribute $u(u - 1)$ such incidences in total.

We need the following coloured-graph observation, setting $K = \max\{12, u + 4\}$. In a simple graph on u vertices, colour each edge with one of at most three colours. If two disjoint edges never have

different colours, the sum of the three colour support sizes is at most K . Here a colour support consists of the vertices incident with edges of that colour.

To prove the observation, suppose first that colour one contains two disjoint edges e, f , and put $W = e \cup f$. Every edge of the other colours must meet both, so its endpoints belong to W . If no other colour is used, the assertion holds. If their combined edge family has no common vertex, each colour-one edge must also have both endpoints in W : an edge with an endpoint outside W would force its other endpoint to be common to that family. All three support sizes are then at most four, giving twelve. Otherwise the combined family has a common vertex $c \in W$. Its edges must join c to one of the two endpoints of the edge among e, f not containing c . There are at most two such edges, and simplicity makes their combined colour-support sum at most four. Adding the first support gives $u + 4$.

Now assume each colour class is pairwise intersecting. A simple pairwise-intersecting graph is a star or is contained in a triangle: take two edges ab, ac ; any edge avoiding a must be bc , which in turn forces every edge into that triangle. If all supports have size at most four, their sum is at most twelve. Otherwise one class is a star with at least four leaves. Every edge of the other colours must contain its centre, since two endpoints cannot meet four distinct leaves. Simplicity ensures each noncentral vertex belongs to at most one support. The centre belongs to at most three, giving a sum at most $u + 2$. This proves the observation.

For each matching triple M_i , form a coloured graph on U : the selected pair xy is given colour $z \in M_i$ when its witness block is $\{x, y, z\}$. Two disjoint pairs with different colours would allow replacing M_i by two disjoint three-blocks, increasing the matching. Thus the observation applies. The colour support sum counts the distinct pairs between U and M_i covered by selected witnesses, so at most K such pairs are covered. Across the s triples, at least $3su - sK$ of the U - C pairs are uncovered by selected witnesses.

For a nonselected vertex let a count its block points in U and c those in C . Since $a + c \leq 3$, one has $ac \leq 2a$. All remaining pairs have nonselected witnesses, so their incidence contribution to U is at least $(3su - sK)/2$. Counting the total ur incidences with U yields

$$ur \geq u(u - 1) + \frac{3su - sK}{2}. \quad (18)$$

This reasoning remains valid if the displayed lower bound is negative.

Equation (16) with $r = u + s - 2 + \alpha$ gives $s \leq u - 3 + 2\alpha$. Dividing (18) by $u > 0$ gives

$$\alpha \geq 1 - \frac{(K - u)s}{2u}. \quad (19)$$

If $\alpha < 0$, its integrality implies $\alpha \leq -1$ and $0 \leq s \leq u - 5$, hence $u \geq 5$. As $K - u \geq 0$, (19) implies $\alpha \geq 1 - (K - u)(u - 5)/(2u)$. This is greater than -1 . Indeed, for $5 \leq u \leq 8$ one has $K = 12$ and

$$4u - (12 - u)(u - 5) = u^2 - 13u + 60 = (u - 13/2)^2 + 71/4 > 0.$$

For $u \geq 8$, one has $K = u + 4$, giving $-1 + 10/u > -1$. This contradiction proves $\alpha \geq 0$, hence (17). Finally

$$4\tau(L) \leq 4s + 4\lceil u/2 \rceil \leq 4s + 2u + 2 \leq q + r + 4.$$

The lemma follows. $\square$

References and provenance

[ABW] Ron Aharoni, János Barát and Ian M. Wanless, *Multipartite hypergraphs achieving equality in Ryser’s conjecture*, *Graphs and Combinatorics* **32** (2016), 1–15, DOI 10.1007/s00373-015-1575-9, arXiv:1409.4833v2. In particular, Theorem 2.5 and Corollary 2.6 supply the comparison bound.

[S] Varun Sivashankar, *An Improved Lower Bound for the Erdős–Lovász Cover Number Problem*, arXiv:2606.24878v2, 29 July 2026. Lemma 2 supplies the degree-three residual inequality and its proof. Theorem 1(i) states $g(r) \geq 3r - 4$ for the different, nonpartite class with $\tau = r$. Its asymptotic Theorem 1(ii) uses an additional edge-colouring theorem; neither that theorem nor its formal assumptions are needed here.

The dated repository literature report records the limited search scope. No OpenAI/math theorem is needed for either theorem of this note. The new partite estimate, its independent consequence and the stronger combination are written explicitly. Lemma 4 and the appendix identify the inherited estimates and proofs separately.